



Gamma rays incoming

The centre of the Milky Way has been sending a lot of gamma rays towards us and dark matter is responsible, well partially. <http://dgit.in/Drkmttgram>



Contact after 13 years

NASA has made contact with a satellite that was lost in orbit 13 years ago and discovered by an amateur astronomer. <http://dgit.in/IMAGESat13>

Although there's debate between marine scientists as to exactly how reefs should be classified, there are mainly five types:

- **Fringing reef:** directly attached to a shore, or borders it with an intervening narrow, shallow channel or lagoon.
- **Barrier reef:** Separated from a mainland or island shore by a deep channel or lagoon.
- **Platform reef:** also called bank or table reefs, which form on the continental shelf, as well as in the open ocean, is the seabed rises close enough to the surface of the ocean.
- **Atoll reef:** more or less circular or continuous barrier reef extends all the way around a lagoon without a central island.

DARWIN'S PARADOX

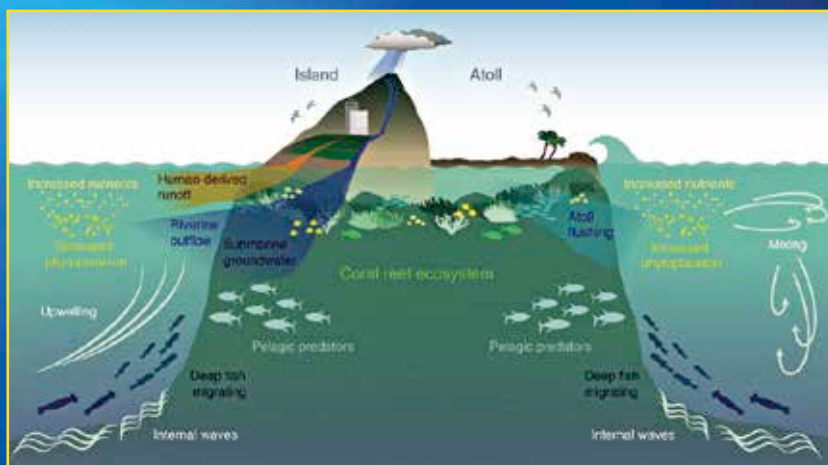
There's a particular characteristic of coral reefs that is paradoxical - their very occurrence. First theorized by Charles Darwin in *The Structure and Distribution of Coral Reefs* in 1842, this



The Great Barrier Reef is the world's largest coral reef system composed of over 2,900 individual reefs and 900 islands stretching for over 2,300 kilometres over an area of approximately 344,400 km²

paradox refers to the unlikely proliferation of coral reefs in tropical waters which contain few nutrients and conditions for their growth. In fact, the largest and strongest corals grew in parts of the reef exposed to the most violent surf and corals were weakened or absent where loose sediment accumulated.

So how do these oasis of oceanic diversity come to be in the middle of aquatic deserts? Scientists explain it with something called the Island Mass



The factors involved in Island Mass Effect (IME)

Effect (IME) that describes a variety of factors that come together to create ideal conditions for coral reef formation.

Geology and bathymetry determine the structure of the sea floor and contribute in determining how much of it receives sunlight and is shallow water. Upwelling brings nutrient rich water from the depths to the sunlight zone. Birds and fish contribute through their waste material. Even human activities like runoff from a nearby agricultural area can affect the IME.

CORALS AT RISK

These beautiful aquatic ecosystems, as mentioned in the beginning, are at severe risk. And it's not just the usual CO₂ to blame. Yes, excessive CO₂ emissions have led to acidification of the oceans, which in turn hinders the formation of the calcium carbonate essential for coral reefs... but that's not all.

Fishing also affects corals. Methods like blast fishing (using explosives) or cyanide fishing (using cyanide to make fish easier to catch) can destroy a thousand year old complex reef ecosystem in minutes. The irony is, it is the same fishing industry, along with tourism and other areas, that comprise the \$30bn+ global reef economy who are now tasked with reef conservation! Even if you don't care about fish or the biodiversity of our oceans, the protection from storms that reefs provide coastal cities, once missing, can lead to devastating consequences!

HOW TO PROTECT CORAL REEFS

Most coral reefs are dying because of rising temperature, which makes them eject the symbiotic algae, and eventually kills them. There are efforts to save them: in some regions, people are setting up sunshades to reduce the impact of solar heating. Cold water from deeper oceans is being pumped into the reefs, which also helps in nutrient cycling.



Blast fishing is the (often illegal) practice of using explosives to stun or kill schools of fish for easy collection.

Scientists are leaving no stone unturned to save our reefs, even resorting to gene editing to identify and inhibit thermal vulnerability in corals. The CRISPR method is being trialled, in the hope that we'd be able to engineer coral that's resistant to higher temperatures. Instead of it happening naturally, we're trying to indulge in some artificial selection. Why? Because when the oceans start dying off, there's a good chance we will go with them! Weather changes with climate change, and we really don't want tsunamis wiping us out! 